Nico Fuchs

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PROFILE

Mechanical-focused engineering student (BSc Human Factors and Sports Engineering, 5th semester, UAS Technikum Wien) with hands-on experience in structural analysis, lightweight and composite design, simulation and sensor-based measurement. Member of TU Wien Racing (Chassis & Composites). Seeking a mandatory bachelor internship in Race-, Automotive- or Composite engineering from February 2027 for 3 to 6 months.

EXPERIENCE

TU Wien Racing – Chassis & Composites Manufacturing Member October 2026 – present
Formula Student Team, Vienna

- Carbon Monocoque construction and Steering wheel prototyping
- CFRP Laminate and tooling construction, 3D Printing

Vienna Event Crew – Event Technician May 2025 – May 2026

- Set up and dismantled stages and event technology in international teams under tight deadlines.
- Troubleshot and resolved technical issues before and during live events.

Compulsory Military Service, Austria 2024

EDUCATION

BSc Sports Engineering and Ergonomics – UAS Technikum Wien 2024 – expected 2027
Curriculum

Relevant coursework: Composite and Lightweight Construction, Strength and Deformation Simulation, Kinematic Simulation of Multi-Body Designs, Machine Elements, Fundamentals of Statics and Dynamics, Material Selection and Manufacturing Processes, Project Management

GRG II Zirkusgasse, Vienna – Matura (Descriptive Geometry) 2023
Final research paper (VWA, grade "very good"): "The Future of Alternative Powertrains in the Automotive Industry" – comparison of electromobility, hydrogen and e-fuels incl. CO2 life cycle assessment.

SKILLS

CAD / FEM Simulation: PTC Creo, Catia V5, 3DExperience, Hyperworks
Programming / Data: Python (NumPy, SciPy, PyTorch), MATLAB, R, Java, JavaScript, Git
3D Printing: FDM, Resin, Blueprint Studio, Chitubox, OrcaSlicer, Bambu Studio
Other software: MS Office, Affinity, Krita
Languages: German, English (C1), Spanish (B1)
Interests: Motorsport, powerlifting, carpentry, motorcycle maintenance, sports photography (club volunteer)

PROJECTS

Structural Analysis of Barbells – Bending and Deflection Study 2026
UAS Technikum Wien, research project

- Non-destructive deflection measurements on barbells under loads of 70–425 kg using 2D photo analysis with automated marker tracking.
- Processed and structured the measurement data for a scientific publication (planned for Winter 2026/27).

AI-Based Impact Localization on a Roundnet – with Spikeball™
Group project (3 students), UAS Technikum Wien, sponsored by Spikeball 2026

- Built a measurement setup with 5 piezoelectric sensors on the frame and an NI USB-6210 DAQ (LabVIEW, 50 kHz); extracted 35 features incl. GCC-PHAT time-difference-of-arrival.
- Compared k-NN, MLP and 1D-CNN (PyTorch) on 200 training and 150 test impacts; the 1D-CNN reached a mean error of 9.4 cm on the net (16.4 cm incl. rim hits) and outperformed the MLP significantly ($p < 0.001$).

Backpack Redesign for Martini Sportswear (Arouca Backpack) 2026
Group project (3 students), UAS Technikum Wien, with industry partner Martini Sportswear

- Analysed competitors (Dynafit, Mammut, DB Journey, USWE) and a biomechanical load study; developed three concepts: material substitution (UHMWPE ripstop, 30–49 % lighter fabric), slimmer and taller shape for better load distribution, and a racing-harness-style strap system.
- Combined the strongest features into one ensemble concept, selected via a weighted decision matrix (9.3 vs. 5.7 for the baseline); planned validation by durability testing, EMG of core and back muscles, and IMU swing tests on an incline treadmill.

SunCafe – Sun/Shade Estimation Web App for Cafés 2026
Personal project, GitHub

- Web app (JavaScript, MapLibre GL, SunCalc) that estimates sun or shade for nearby cafés up to 48 h ahead, incl. building shadows and opening hours.
- Python pipeline generating terrain/building horizon masks (180 sectors) from open data, automated with GitHub Actions; serverless backend on Vercel.
- Unit tests (Node, pytest), ESLint and CI on every push; 200+ commits.